

Xintong Li (李欣桐)

Email: lixt21@mails.tsinghua.edu.cn | Telephone/Wechat: +86-18210606882

Education

Tsinghua University

Sept. 2021 – Present

- Institute for Interdisciplinary Information Sciences (IIIS)
- Ph.D. student in Computer Science, advised by Prof. Mingyu Gao (高鸣宇).

Peking University

Sept. 2017 – June 2021

- School of Computer Science
- B.S. in Computer Science, advised by Prof. Guangyu Sun (孙广宇).

Research Interests

I specialize in computational acceleration, with a particular focus on accelerating sparse computations. This involves analyzing the impact of different sparse features on computational aspects such as tiling; scheduling; buffer management, and optimizing for specific sparse characteristics. Additionally, I am interested in exploring other computational models, such as near-data processing (NDP).

Leading Projects

[ISCA 2025] HYTE: Flexible Tiling for Sparse Accelerators via Hybrid Static-Dynamic Approaches

- Research Problem: Sparse features significantly impact the effectiveness and overhead of tiling in sparse tensor algebra. The optimal tile size; tile shape; inter-tile order; and buffer allocation vary across different sparse matrices. Additional hardware support is also required to explore the full design space.
- Define the full design space of sparse tiling and explore the influences of different sparse features.
- Propose a static offline scheduler to identify a near-optimal initial tiling scheme through effective and lightweight sampling.
- Supports runtime efficient management of the tiling metadata in both the off-chip memory and the on-chip buffer, as well as a technique of dynamic tuning on the tile shape to ensure high buffer utilization in the presence of highly varying data patterns.
- Outperforms state-of-the-art sparse tiling strategies by $3.9\times$ to $7.4\times$ on average for diverse sparse matrices.

[MICRO 2025] SeaCache: Efficient and Adaptive Caching for Sparse Accelerators

- Research Problem: Sparse accelerators introduce new features, including variable fiber length and distribution; the need for random indexing and eviction; and irregular yet predictable access patterns. Conventional cache designs in general-purpose hardware or dense accelerators are unable to accommodate these new features.
- Design a two-part address mapping method to avoid cacheline underutilization or high indirect access costs.
- Consider the trade-off between prefetching costs and hit rate improvements, and design a prefetch data management strategy with lower costs. (Sparse computations can leverage prefetching from one matrix to predict the future access patterns of another matrix, thereby achieving better replacement and improving cache hit rates.)
- Explore the optimal buffer allocation ratio among the two-part data, metadata, and prefetch data.

[In Progress] AutoNDP: Automating Computation Scheduling in Near-Data Processing Systems for LLM Inference

- Develop a hybrid mapping strategy combining compute-centric mapping (mapping operators to computational units, which leads to DRAM bank capacity imbalance and additional communication in NMP scenarios) and data-centric mapping (mapping data such as weights and KV cache to DRAM banks, which causes computational unit imbalance). The specific implementation dynamically maps each tile (divided operators) based on runtime conditions.
- During simulation, store the performance of operators with specific shapes on specific hardware, simulating operators of the same shape only once. This avoids the inaccuracy of analytical models and the slow speed of

cycle-accurate models.

- In implementation, incorporate tile size and shape selection for NMP scenarios, allowing the same operator to be stored/computed in different locations; dynamically allocated tiles based on runtime conditions; enhanced precise simulation and evaluation of softmax; used a simulator integrating the above three components for hardware Design Space Exploration (DSE).

Experience

Heterogeneous Computing Group, ByteDance

April 2025 - Present

- Evaluate and analyze the performance of new hardware technologies, such as PIM and RRAM, on business models, analyzing the gap between theoretical and actual implementations.
- Complete the autoNDP project.

Publications

- **SeaCache: Efficient and Adaptive Caching for Sparse Accelerators** MICRO, 2025
Xintong Li, Jincheng Jiang, Mingyu Gao
- **HYTE: Flexible Tiling for Sparse Accelerators via Hybrid Static-Dynamic Approaches** ISCA, 2025
Xintong Li, Zhiyao Li, Mingyu Gao
- **Adyna: Accelerating Dynamic Neural Networks with Adaptive Scheduling** HPCA, 2025
Zhiyao Li, Bohan Yang, Jiayang Li, Taijie Chen, *Xintong Li*, Mingyu Gao
- **Hardware-Assisted Service Live Migration in Resource-Limited Edge Computing Systems** DAC, 2020
Zhe Zhou, *Xintong Li*, Xiaoyang Wang, Zheng Liang, Guangyu Sun, Guojie Luo

Honors and Awards

- National Scholarship 2020
- Honored Student 2019, 2020
- National Olympiad in Informatics (NOI), Silver Medal 2015

Teaching

- Teaching Assistant Fall 2023, Fall 2024
20470084 Computer Architecture, Tsinghua University
Maintained and improved a course project focused on a computer architecture simulator for RISC-V.
Taught review session classes on review and expansion.